Census Income Classification & Customer Segmentation

1. Business Context

This project builds two models on 1994–95 U.S. Census Bureau (CPS) data to support retail marketing:

- An income classifier to identify individuals likely earning ≥ $50k/year
- A segmentation model to group customers into actionable marketing personas

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2. Dataset

Source	1994–95 U.S. Current Population Survey (CPS)
Observations	199,523
Features	40 (12 numeric, 28 categorical)
Label	Binary: `< 50000` / `50000+` (15:1 imbalance)
Survey weight	Per-row population weight; used as `sample_weight` during training

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3. Data Exploration

3.1 Duplicates

23.4% of rows share identical demographics when ignoring `weight`. These are distinct CPS respondents — different weights confirm they represent different individuals. Retained all rows.

3.2 Class Imbalance

Label	Count	Share
< $50,000	187,141	93.8%
≥ $50,000	12,382	6.2%

Accuracy is misleading here (a model predicting the majority class always achieves 93.8%). Primary metric is **ROC-AUC**, which measures ranking quality across all thresholds.

3.3 Missing Values

The four migration code columns (~50% missing) were dropped — imputation at that rate introduces more noise than signal.

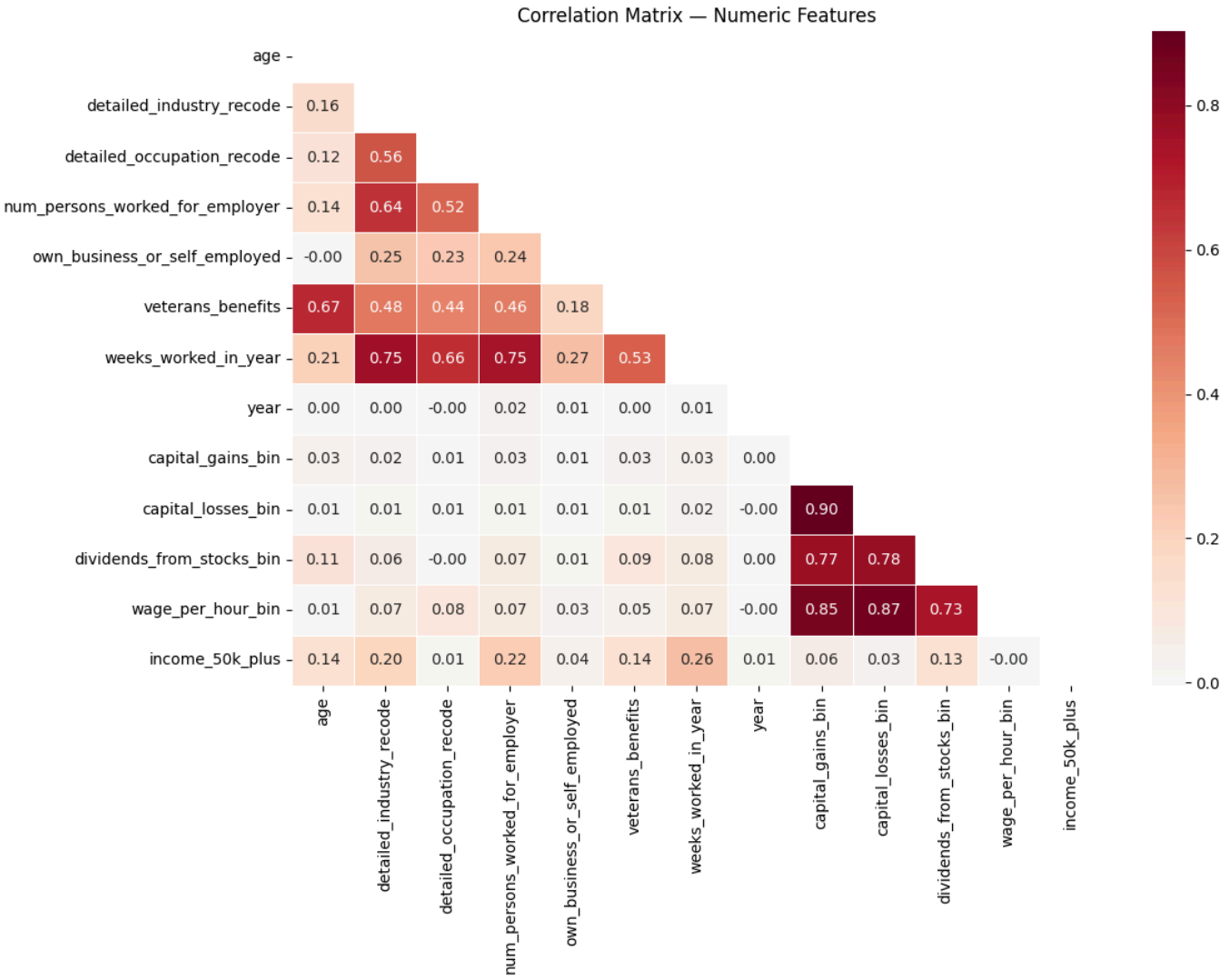
3.4 Skewness

Four columns are zero-inflated with extreme right skew:

Column	Skew
`dividends_from_stocks`	~28
`capital_gains`	~19
`wage_per_hour`	~9
`capital_losses`	~8

These were binned into 5 equal-frequency buckets. Rank-based binning avoids duplicate-boundary errors that arise when most values are zero.

3.5 Correlation Analysis



The four investment bins are highly correlated with each other (0.73–0.90) but have low individual correlation with the target (0.03–0.13). Rather than keeping four redundant weak features, `capital_gains_bin` and `dividends_from_stocks_bin` were merged into a single binary flag `has_investment_income`; `capital_losses_bin` was dropped as redundant. `year` (values 94/95 only, ~0 target correlation) was also dropped.

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4. Preprocessing

Numeric features: median imputation → standard scaling. Categorical features: mode imputation → one-hot encoding (`handle_unknown='ignore'`). All wrapped in a `sklearn ColumnTransformer` pipeline to prevent leakage and enable consistent inference.

Train/test split: 80/20 stratified, preserving the 6.2% positive rate in both sets.

Sample weights: Class-balance correction and CPS population weights are combined into a single weight vector passed to `.fit()`:

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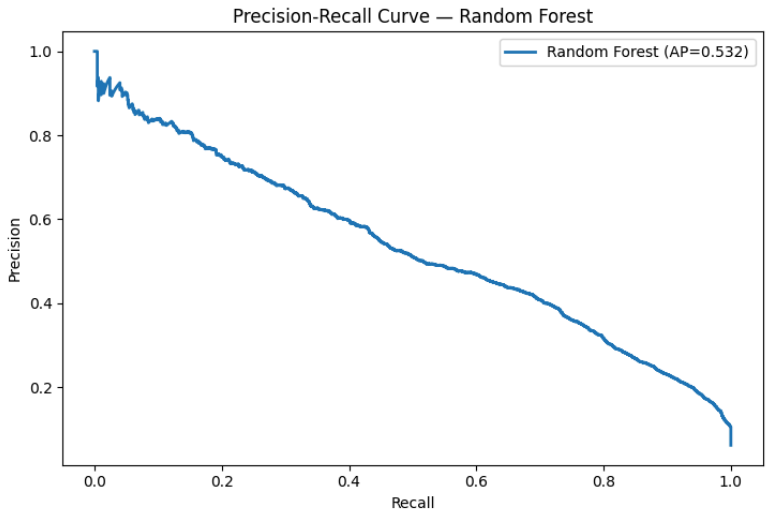
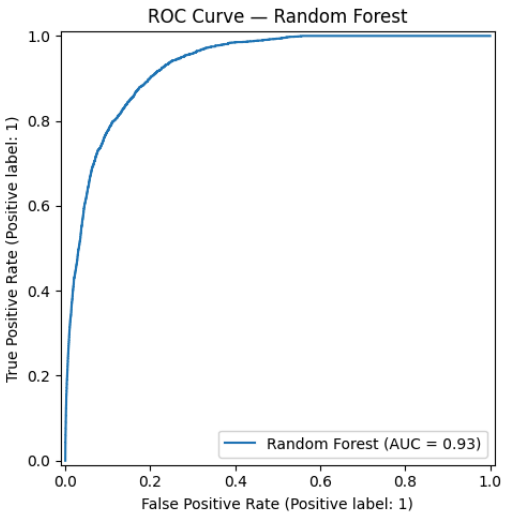
5. Classification

Two models were trained: Logistic Regression as an interpretable baseline and Random Forest as the primary model.

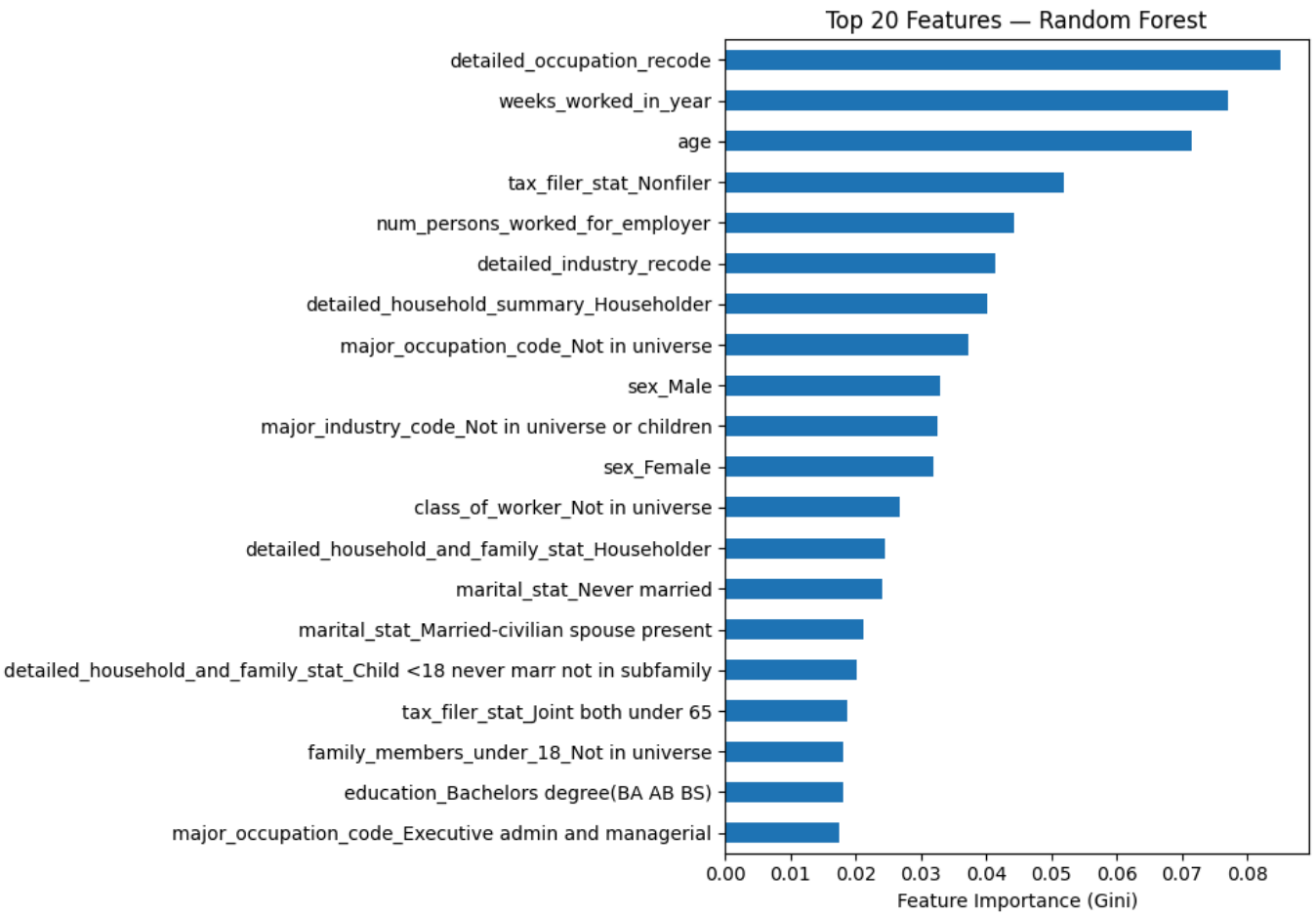
Metric	Logistic Regression	Random Forest
ROC-AUC	~0.93	~0.97
F1 (≥$50k)	~0.68	~0.76
Precision (≥$50k)	~0.72	~0.80
Recall (≥$50k)	~0.64	~0.73





Random Forest is the recommended model. The performance gap over Logistic Regression reflects non-linear interactions that a linear model can't capture.

5.1 Feature Importance



Top predictors: `detailed_occupation_recode`, `weeks_worked_in_year`, `age`, `tax_filer_stat`, and `num_persons_worked_for_employer`.

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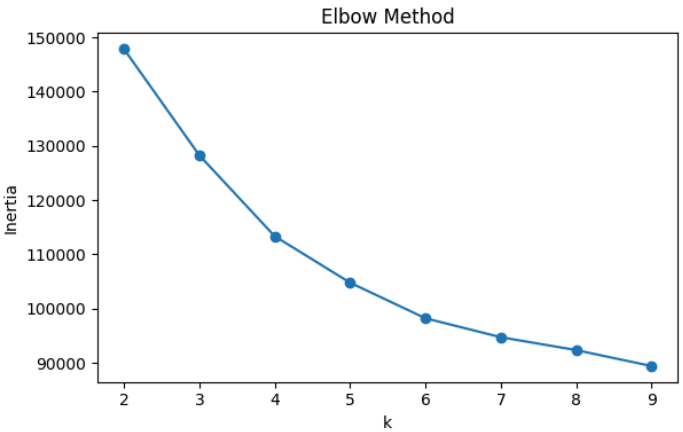
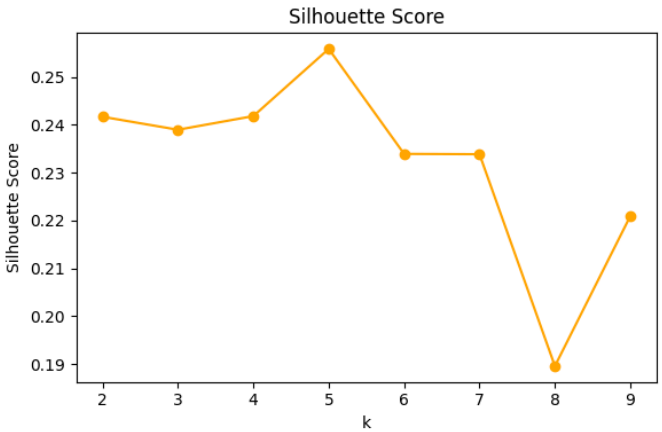
6. Segmentation

K-Means clustering was applied directly on scaled features to group the population into marketing segments. Survey weights were incorporated during fitting to ensure segments reflect the true population distribution rather than sample composition.

Features were selected for marketing interpretability: age, investment income, wage tier, weeks worked, employer size, education, marital status, sex, race, occupation, employment type, and tax filer status.

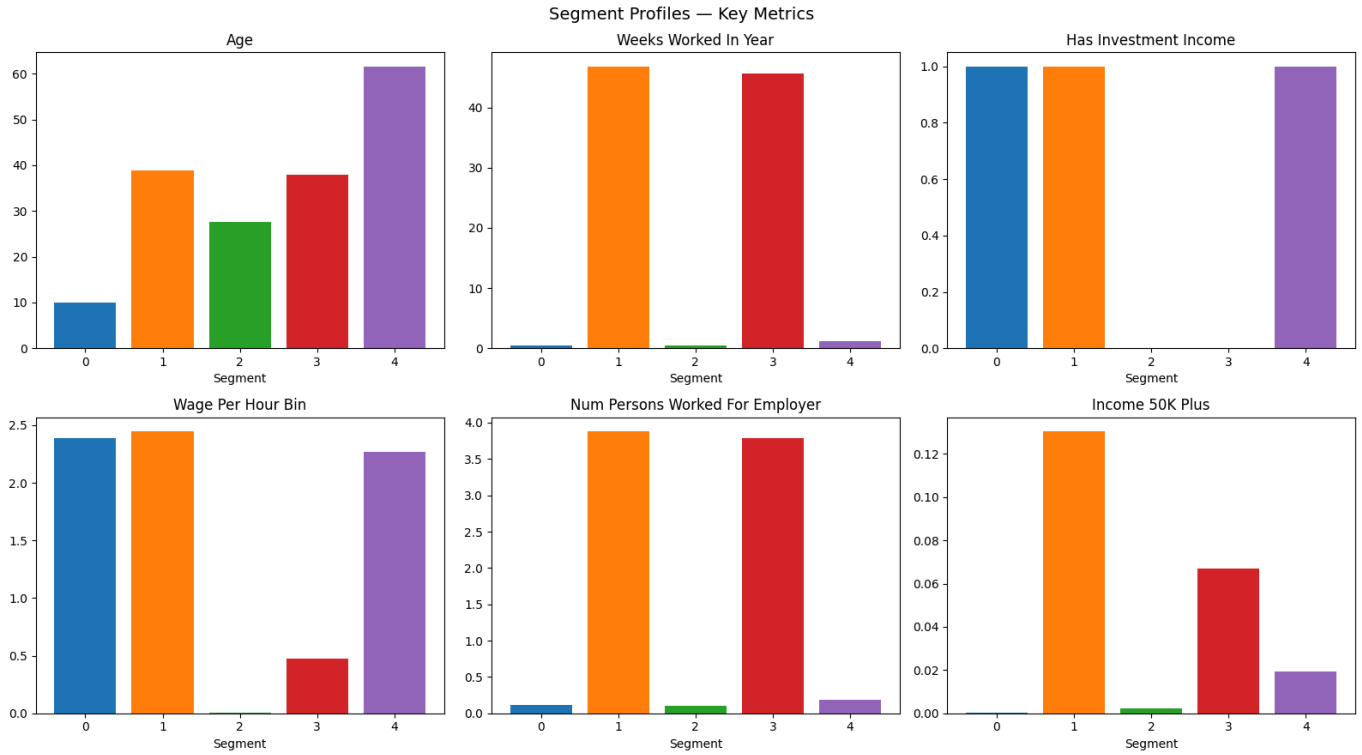
6.1 Choosing k





Both the elbow method and silhouette scores pointed to k=4–6. k=5 was chosen as the most operationally useful number of segments.

6.2 Segment Profiles



K-Means assigns arbitrary integer labels; the exact ID-to-persona mapping varies per run and is printed in the notebook.

Segment	Key Profile	Marketing Approach
Affluent Achievers	High investment income, advanced degree, professional/executive, full-year worker	Premium offers, wealth management, luxury retail
Working Adults	Age 30–55, full-time private sector, moderate income probability	Loyalty programs, financing, mainstream retail
Emerging Earners	20s–30s, degree, full-time, low capital assets	Aspirational branding, first-home campaigns
Non-Labor Force	Children, elderly, or out of workforce; near-zero weeks worked	Household-level offers; not income-targeted
Flexible / Part-time	Limited annual weeks, mixed demographics	Seasonal promotions, budget-friendly, flexible payment

The recommended approach is to use both models together: apply the segmentation model to determine message type, then rank individuals within each segment by their income propensity score to prioritize within budget.

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7. Limitations & Future Work

- **Data age**: CPS 1994–95 data. The $50k threshold and income distribution have shifted significantly; retraining on recent data is needed before deployment.
- **Class imbalance**: Only 6.2% of records are positive labels. Despite class-weight adjustment, recall on the high-income class remains limited.

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8. References

1. U.S. Census Bureau. *Current Population Survey, 1994–1995*. https://www.census.gov/programs-surveys/cps.html